

LOUVAIN SCHOOL OF MANAGEMENT

Group Project Report

LLSMS2138 - Big Data in Finance

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Date:

May 22, 2026

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0 Executive Summary

1 In-Sample and Out-of-Sample Performance

The primary objective of this section is to evaluate the in-sample (IS) and out-of-sample (OOS) performance of three foundational portfolio allocation strategies. By applying these strategies across four diverse sets of equity portfolios sourced from the Kenneth French Data Library (10 industry portfolios, 25 portfolios formed on size and book-to-market, 48 industry portfolios, and 100 portfolios formed on size and book-to-market), we aim to empirically demonstrate how dimensionality, return frequency, and estimation error impact portfolio stability and actual risk-adjusted returns.

To formalize the allocation problem, let R_t denote the $N \times 1$ vector of raw returns for a universe of N assets at time t . Throughout this analysis, we operate under the assumption that the risk-free rate is zero, which allows us to mathematically treat these raw returns directly as excess returns. Let w represent the $N \times 1$ vector of portfolio weights. Our evaluation benchmarks a naive diversification approach against two classic variance-based optimization frameworks.

1.1 Equally Weighted (1/N) Portfolio

This naïve diversification strategy assigns equal weight to all assets in the universe, completely bypassing the need for parameter estimation. The weight vector is defined strictly as:

$$w_{1/N} = \frac{1}{N} \iota$$

where ι is an $N \times 1$ vector of ones.

Despite its mathematical simplicity, the $1/N$ portfolio serves as a highly robust benchmark. Its primary advantage is that it is not prone to estimation error because it does not rely on sample moments. Therefore, it cannot be deceived by historical noise. This structural robustness often allows it to remain highly competitive out-of-sample, particularly in high-dimensional environments where optimized models break down. That being said, this resilience comes at the cost of theoretical efficiency. By completely ignoring the asset covariance structure ($\hat{\Sigma}$) and expected returns ($\hat{\mu}$), the strategy fails to properly allocate risk or optimally exploit diversification benefits. Consequently, it consistently plots inside the efficient frontier, visibly underperforming optimized frameworks in an in-sample setting.

1.2 Sample Minimum-Variance Portfolio (MVP)

This strategy seeks to minimize portfolio variance without relying on expected return estimates. Using the sample covariance matrix $\hat{\Sigma}$, the optimal weights are derived by solving the optimization problem $\min w^T \hat{\Sigma} w$ subject to the fully invested constraint $w^T \iota = 1$. The closed-form analytical solution is:

$$w_{mvp} = \frac{\hat{\Sigma}^{-1} \iota}{\iota^T \hat{\Sigma}^{-1} \iota}$$

The primary theoretical advantage of the MVP is its independence from the expected return vector ($\hat{\mu}$), which is notoriously difficult to estimate and highly susceptible to historical noise. By relying solely on the covariance structure, the MVP generally exhibits superior out-of-sample stability compared to frameworks that attempt to maximize returns. Despite this advantage, this strategy is still vulnerable to estimation error because it relies entirely on the inverse of the sample covariance matrix ($\hat{\Sigma}^{-1}$). In high-dimensional settings—where the number of assets N is large relative to the number of historical observations T , estimating the covariance between every single pair of assets becomes highly unreliable. When the optimizer mathematically inverts this noisy matrix, it essentially magnifies the underlying estimation errors. This leads to unstable portfolio allocations characterized by extreme and unrealistic long and short positions.

1.3 Sample Mean-Variance Portfolio (Max Sharpe)

This strategy, rooted in classical Markowitz optimization, seeks to maximize the risk-adjusted return (the Sharpe ratio). By utilizing both the sample mean vector of expected returns $\hat{\mu}$ and the sample covariance matrix $\hat{\Sigma}$, the optimal weights are determined by:

$$w_{mv} = \frac{\hat{\Sigma}^{-1}\hat{\mu}}{\iota^T \hat{\Sigma}^{-1}\hat{\mu}}$$

Theoretically, the Mean-Variance portfolio represents the absolute optimal allocation. In an in-sample setting where historical data is treated as perfect information, this strategy mathematically dominates all others by identifying the precise tangency portfolio on the efficient frontier. The critical flaw here is that the Mean-Variance optimizer is extraordinarily sensitive to estimation error because historical mean returns ($\hat{\mu}$) are incredibly noisy and unreliable predictors of future returns. It aggressively overweights assets that experienced lucky, spuriously high past returns and heavily shorts those with unlucky past returns. When this flawed $\hat{\mu}$ vector is multiplied by the inverted covariance matrix ($\hat{\Sigma}^{-1}$), the statistical noise compounds. This combination of errors leads to the catastrophic weight explosions and massive performance degradation observed in high-dimensional datasets.

1.4 Evaluation Framework: In-Sample vs. Out-of-Sample

To establish a rigorous comparison, we impose a fully invested constraint ($w^T \iota = 1$) across all strategies while permitting unconstrained short-selling, allowing the optimizers to freely utilize leverage via negative weights. For the in-sample (IS) analysis, we compute a single, static set of portfolio weights utilizing the entire dataset. By providing the optimizer with perfect foresight over the full sample to estimate $\hat{\mu}$ and $\hat{\Sigma}$, this approach yields a theoretical upper bound for risk-adjusted returns, establishing a baseline against which we can measure the impact of estimation error.

To evaluate realistic, achievable performance, we conduct an out-of-sample (OOS) analysis using a rolling-window methodology. We define a backward-looking estimation window, W , of exactly 10 years (120 months or 2,520 trading days) and a forward-looking holding period, H , of 6 months (126 trading days). At any evaluation time t , we calculate the sample moments ($\hat{\mu}_t$ and $\hat{\Sigma}_t$) using only the observable historical returns within the estimation window $[t - W + 1, t]$. We then compute the optimal weight vector w_t and

hold this allocation fixed over the subsequent test window $[t + 1, t + H]$. The realized OOS portfolio return for any period $t + h$ within this holding window is mathematically defined as:

$$r_{p,t+h} = w_t^T R_{t+h}$$

The estimation window is then rolled forward by H periods, and this process iterates until the end of the dataset is reached. Both the IS and OOS frameworks are ultimately evaluated across three standardized metrics. First, the annualized expected return is computed as $\hat{\mu}_p \times \text{freq}$, where the multiplier is 12 for monthly data and 252 for daily data. Next, the annualized volatility (risk) is computed as $\hat{\sigma}_p \times \sqrt{\text{freq}}$. Lastly, the Sharpe ratio is calculated as the ratio of the annualized return to the annualized volatility, reflecting the premium earned per unit of risk under our zero risk-free rate assumption.

1.5 Empirical Results: IS vs. OOS

Table 1 presents the comprehensive performance metrics, annualized mean return, volatility, and Sharpe ratio, for all three allocation strategies evaluated across the four equity datasets and two data sampling frequencies.

Dataset	Strategy	Monthly Frequency			Daily Frequency		
		Mean	Vol	Sharpe	Mean	Vol	Sharpe
10 Industry	IS: 1/N	0.1210	0.1504	0.8048	0.1217	0.1601	0.7599
	IS: Min-Variance	0.1161	0.1233	0.9421	0.1156	0.1320	0.8754
	IS: Mean-Variance	0.1236	0.1272	0.9720	0.1230	0.1362	0.9032
	OOS: 1/N	0.1312	0.1470	0.8925	0.1333	0.1662	0.8023
	OOS: Min-Variance	0.1217	0.1240	0.9816	0.1224	0.1340	0.9130
	OOS: Mean-Variance	0.1199	0.1715	0.6995	0.1124	0.2155	0.5215
25 Portfolios	IS: 1/N	0.1315	0.1822	0.7218	0.1291	0.1748	0.7387
	IS: Min-Variance	0.1497	0.1232	1.2151	0.1601	0.1293	1.2381
	IS: Mean-Variance	0.2857	0.1702	1.6789	0.3288	0.1853	1.7742
	OOS: 1/N	0.1355	0.1762	0.7688	0.1359	0.1830	0.7425
	OOS: Min-Variance	0.1791	0.1355	1.3219	0.1987	0.1316	1.5099
	OOS: Mean-Variance	0.2601	0.2523	1.0309	0.2771	0.2069	1.3396
48 Industry	IS: 1/N	0.1218	0.1706	0.7141	0.1208	0.1669	0.7238
	IS: Min-Variance	0.1062	0.1103	0.9627	0.1057	0.1209	0.8743
	IS: Mean-Variance	0.2354	0.1642	1.4336	0.2331	0.1796	1.2982
	OOS: 1/N	0.1303	0.1640	0.7947	0.1319	0.1722	0.7659
	OOS: Min-Variance	0.0997	0.1423	0.7006	0.1111	0.1241	0.8953
	OOS: Mean-Variance	0.1449	0.4232	0.3423	0.1206	0.3600	0.3350
100 Portfolios	IS: 1/N	0.1322	0.1842	0.7179	0.1299	0.1750	0.7423
	IS: Min-Variance	0.1516	0.1092	1.3876	0.1748	0.1069	1.6350
	IS: Mean-Variance	0.4770	0.1938	2.4615	0.4182	0.1654	2.5288
	OOS: 1/N	0.1363	0.1780	0.7660	0.1367	0.1831	0.7469
	OOS: Min-Variance	0.1814	0.2557	0.7094	0.1951	0.1106	1.7645
	OOS: Mean-Variance	1.5031	15.5141	0.0969	0.3363	0.2340	1.4373

Table 1: Portfolio strategy performance metrics. Mean and volatility are annualized.

In the in-sample (IS) evaluation, the Mean-Variance strategy yields the highest Sharpe ratios across all portfolios. This outperformance is expected due to look-ahead bias. Because the optimizer calculates the weights using the entire 50-year dataset at once, it essentially uses future data. It simply overweights the assets that had the highest returns over the whole period. This overfitting creates a perfect Efficient Frontier (see Figure 1), showing a theoretical maximum rather than a realistic trading strategy. In contrast, the $1/N$ and Minimum-Variance strategies appear mathematically suboptimal here precisely because they do not benefit from this look-ahead bias.

By contrast, the out-of-sample (OOS) results tell a very different story. In a realistic setting, the Mean-Variance portfolio often fails to beat the simple $1/N$ benchmark. This underperformance is caused by estimation error. Because the rolling-window approach uses only past data to predict the future, the optimizer is easily fooled by random noise. It takes unreliable historical averages ($\hat{\mu}$) and covariances ($\hat{\Sigma}$) and aggressively buys assets that simply had good past returns, while heavily shorting the bad ones. Instead of maximizing actual future returns, it ends up maximizing these past errors. This problem gets much worse as the number of assets (N) increases. For example, in the 100-portfolio dataset using monthly returns, the optimizer has to estimate 5,050 covariance parameters from only 120 months of data. Because there are far more parameters to estimate than actual data points, the calculations become highly unstable. This causes the out-of-sample volatility to shoot up to over 15.51 (annualized) and the Sharpe ratio to collapse to just 0.0969.

Conversely, the $1/N$ and Minimum-Variance portfolios demonstrate significant out-of-sample resilience. The $1/N$ strategy consistently holds its ground because it requires zero parameter estimation, completely immunizing it against historical noise. Meanwhile, the Minimum-Variance strategy often delivers the lowest actual OOS volatility. By entirely ignoring the expected return vector ($\hat{\mu}$), which is notoriously difficult to forecast, and focusing strictly on the covariance matrix ($\hat{\Sigma}$), it drastically limits its exposure to estimation error and produces much more stable, realistic allocations over time.

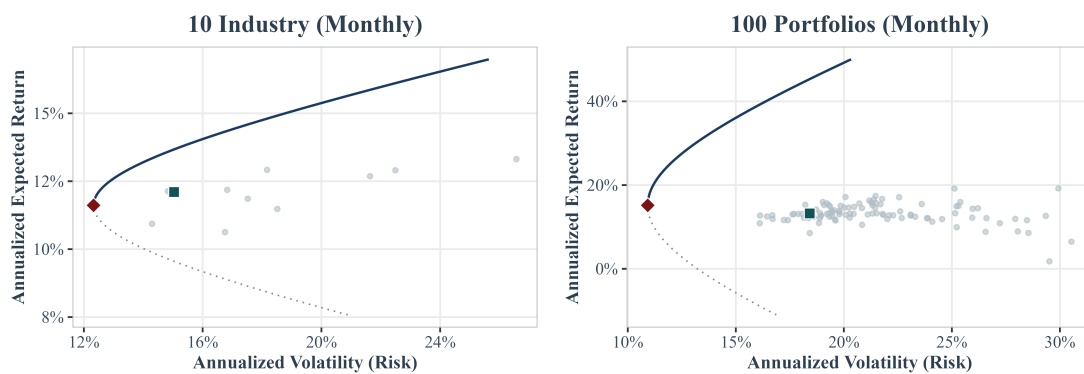


Figure 1: In-sample Efficient Frontiers for the 10-Industry (left) and 100-Portfolio (right) monthly datasets, illustrating the theoretical placement of the Minimum-Variance (diamond) and Equally Weighted strategies (square).

1.6 Stability of Portfolio Weights: Daily vs. Monthly Frequencies

To understand the mechanical drivers of the performance differences between daily and monthly datasets, we must examine how portfolio allocations fluctuate over time. When switching to a daily frequency, optimized strategies generally stabilize significantly compared to their monthly counterparts. With a 10-year rolling window consisting of 2,520 days, the daily asset returns provide a substantially larger sample size to estimate the covariance matrix compared to just 120 monthly observations. This increased sample density minimizes the variance of the estimators, smoothing out statistical noise and tempering the extreme weight fluctuations that structurally break the monthly optimizations.

To visualize this, we extract the rolling weights calculated during the out-of-sample loops. For the 10-Industry dataset, we plot the distribution for each industry individually. For the larger, high-dimensional datasets ($N = 25, 48, 100$), plotting individual assets becomes visually unreadable, so we aggregate the overall distribution of all asset weights across the strategies. We expect the naïve $1/N$ portfolio weights to remain perfectly constant. Conversely, the Mean-Variance framework should exhibit the most instability (especially at monthly frequencies), whereas the Minimum-Variance weights should remain relatively stable, with stability increasing as we switch from monthly to daily observations.

The empirical results for the 10-Industry dataset somewhat challenge our initial intuitions. We assumed that asset weights would fluctuate the most as the observation frequency decreased. However, because the covariance matrix is relatively stable in both frequencies for such a small number of assets ($N = 10$), the daily optimizer actually picks up on high-frequency noise, short-term volatility spikes, and daily price reactions. This causes the optimal weight allocations to adjust dynamically and shift aggressively from day to day, creating wider box distributions and longer whiskers on the daily OOS charts compared to the smoothed allocations computed from monthly OOS data. That said, the Mean-Variance strategy still exhibits the largest structural weight swings across both frequencies as we expected.

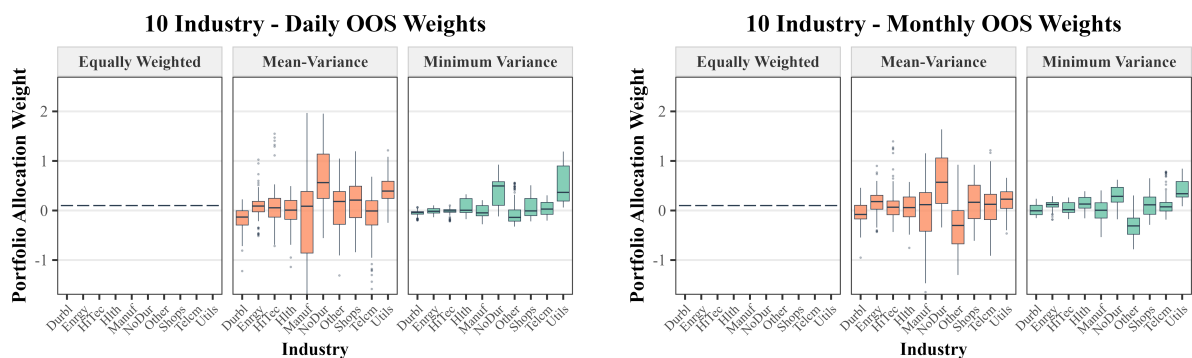


Figure 2: Daily vs. monthly OOS weight distributions for the 10 Industry portfolios.

For the 25 Portfolios based on size and book-to-market, the results perfectly align with our expectations regarding sample size. The Mean-Variance strategy in monthly frequencies is highly unstable, taking aggressive short positions reaching -750% leverage. The daily

observations possess the statistical power to smooth out these extreme fluctuations and this is why the daily Mean-Variance strategy never shorts an asset beyond -300%.

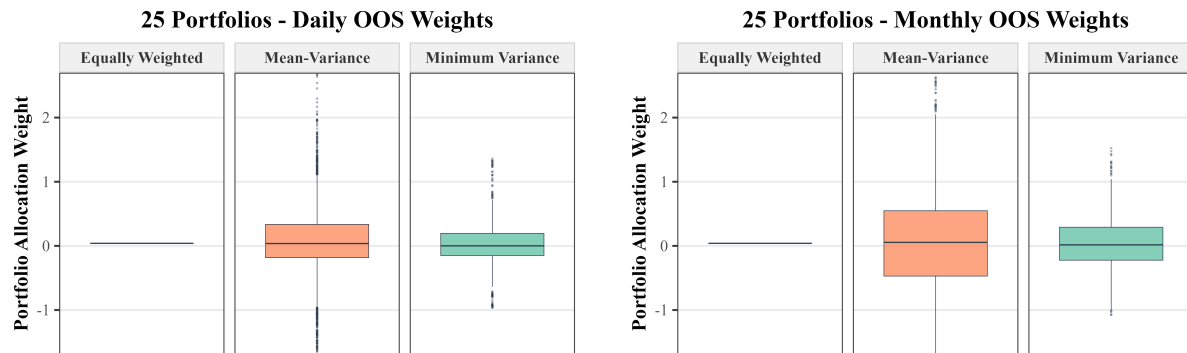


Figure 3: Daily vs. monthly OOS weight distributions for the 25 Portfolios.

The 48 Industry portfolios further confirm the impact of sample size. Daily weights experience much smoother transitions than their monthly counterparts. As expected, the Mean-Variance allocation remains structurally the most volatile strategy across both frequencies due to its strict reliance on the highly sensitive $\hat{\mu}$ vector.

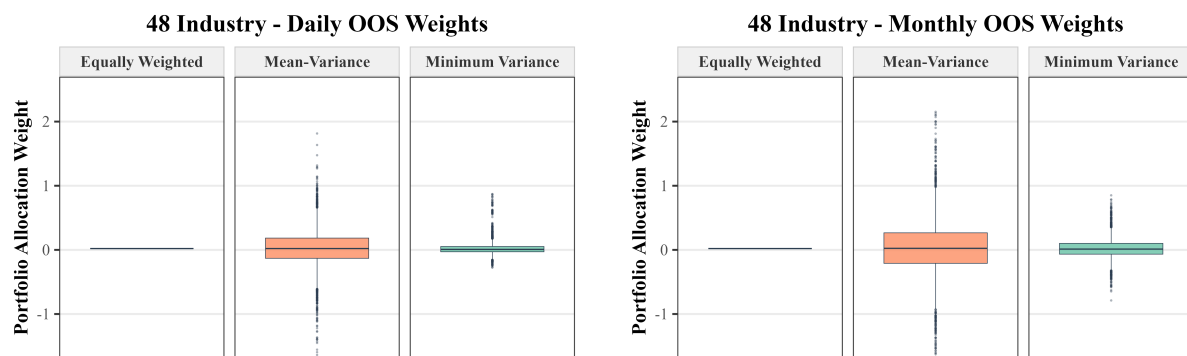


Figure 4: Daily vs. Monthly OOS weight distributions for the 48 Industry portfolios.

The 100 Portfolios based on size and book-to-market dataset most clearly demonstrates the catastrophic effect of estimating too many parameters with too little data. In the monthly frequency, certain Mean-Variance weights explode to 80,000%, balanced by massive short positions reaching -60,000%. In stark contrast, the daily frequency provides enough degrees of freedom to constrain the maximum long exposure to roughly 200%, powerfully demonstrating the critical importance of observation availability when applying unconstrained Mean-Variance optimization.

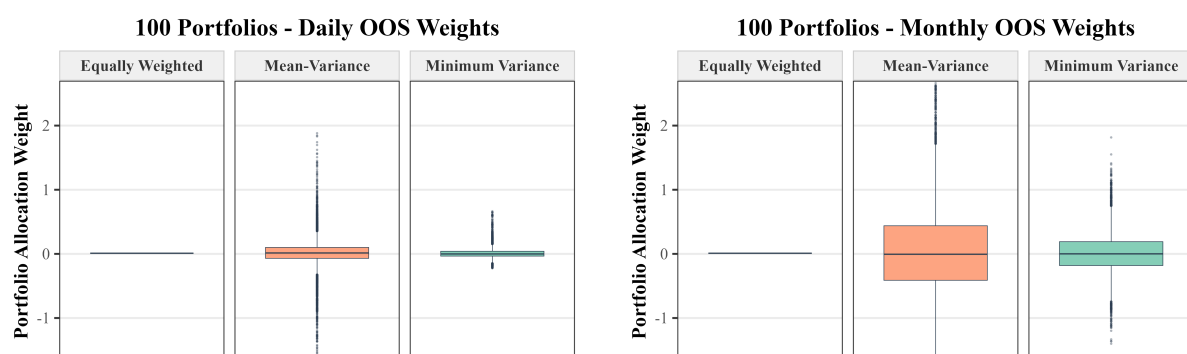


Figure 5: Daily vs. monthly OOS weight distributions for the 100 Portfolios.

2 ℓ_2 -Norm-Constrained Minimum-Variance Portfolio

As we saw in section 1 for the Minimum-Variance portfolio, the sample covariance matrix contains a lot of estimation error because it relies purely on historical data. The unconstrained minimum-variance optimizer reacts too strongly to this noise, producing extreme and unrealistic weights. The ℓ_2 penalty is a direct solution to this problem. The strategy introduces a penalty term based on the sum of the squared weights:

$$w'w = \sum_{i=1}^N w_i^2$$

By squaring the weights, the optimizer is mathematically punished for concentrating capital. A single massive long or short position (for example, 500% in one asset) incurs a massive penalty. Conversely, spreading that same capital evenly across many smaller positions incurs almost no penalty. This naturally discourages extreme leverage, making the portfolio much less sensitive to estimation errors and usually more stable.

This creates a classic bias-variance trade-off. The regularized portfolio is more biased because it is pushed away from the pure unconstrained solution, but it can have much lower estimation variance and better out-of-sample performance. This is especially useful when the number of assets is large relative to the sample size.

2.1 Methodology

Mathematically, the ℓ_2 -regularized minimum-variance portfolio is defined as:

$$\hat{w}_\gamma = \arg \min_{w' \mathbf{1} = 1} (w' \hat{\Sigma} w + \gamma w' w)$$

where $\hat{\Sigma}$ is the sample covariance matrix and $\gamma \geq 0$ is the regularization parameter. γ controls how strictly we enforce the penalty: a higher γ punishes extreme weights more heavily, while a γ of 0 removes the penalty entirely. The closed-form analytical solution to this optimization is:

$$\hat{w}_\gamma = \frac{(\hat{\Sigma} + \gamma I_N)^{-1} \mathbf{1}}{\mathbf{1}'(\hat{\Sigma} + \gamma I_N)^{-1} \mathbf{1}}$$

This reveals that applying an ℓ_2 penalty is equivalent to adding a constant γ strictly to the diagonal elements of the covariance matrix: $\tilde{\Sigma} = \hat{\Sigma} + \gamma I_N$. By doing this, we are essentially adding a small amount of extra variance (risk) to every single asset. To understand why this works, it helps to think about how the optimizer behaves. If historical data randomly makes one asset look incredibly safe, the optimizer will aggressively pour all the portfolio's capital into it. By adding γ to the diagonal, we artificially add a baseline level of risk to every single asset. Because no individual asset looks perfectly safe anymore, the optimizer can no longer go all-in on just one. Instead, it is forced to look for safety the right way: by spreading the money out to benefit from the correlations between different assets. Mathematically, this also fixes the weight explosion. An unstable covariance matrix is like a number dangerously close to zero. When the optimizer tries to invert it, it is essentially trying to divide by zero, which is exactly why the unconstrained weights shot up to 80,000% in section 1. Adding γ gives the matrix a solid mathematical floor, ensuring it can be inverted safely and keeping the allocations grounded.

We maintain the exact same rolling-window methodology established in section 1. For each dataset, we use 10 years of returns to estimate the portfolio, evaluate it over the next six months, and then roll the window forward by six months. Performance is evaluated using annualized mean return, annualized volatility, and the Sharpe ratio.

2.2 Empirical Results: Performance Dependency on γ

The out-of-sample performance heavily depends on the magnitude of the penalty parameter, γ , as well as the underlying sample size of the dataset. Table 2 summarizes the best risk-adjusted performance achieved across all datasets, while Figure 6 illustrates this dynamic visually for both frequencies.

Dataset	Best γ	Ann. Mean	Ann. Vol	Sharpe Ratio
10 Ind Daily	2.30e-05	12.61%	13.53%	0.932
10 Ind Monthly	4.62e-04	12.24%	12.04%	1.016
25 Size Daily	1.46e-08	19.86%	13.16%	1.510
25 Size Monthly	2.25e-05	17.54%	13.08%	1.341
48 Ind Daily	9.21e-05	12.08%	12.73%	0.949
48 Ind Monthly	1.97e-03	12.09%	11.89%	1.016
100 Size Daily	1.72e-08	19.51%	11.06%	1.764
100 Size Monthly	2.12e-04	18.91%	13.44%	1.407

Table 2: Best out-of-sample performance of the ℓ_2 -regularized minimum-variance portfolio for each dataset.

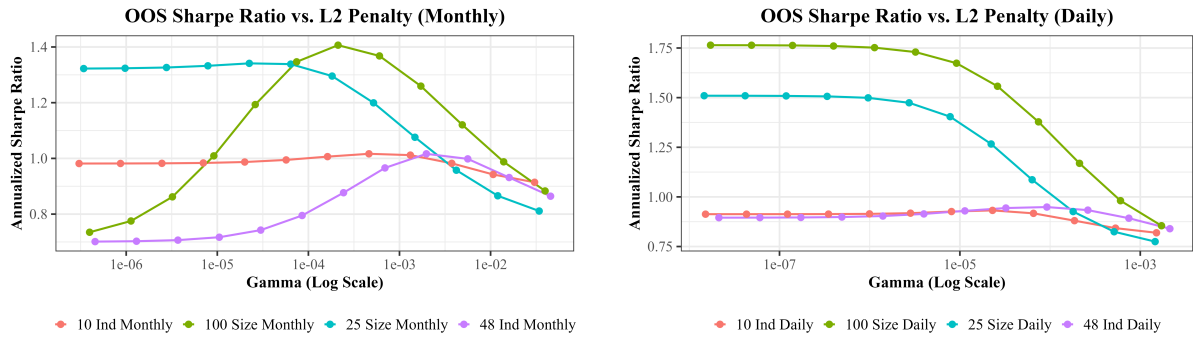


Figure 6: Out-of-sample annualized Sharpe Ratio across a logarithmic grid of γ penalties for Monthly (left) and Daily (right) datasets.

The two charts reveal a striking contrast between the frequencies. For the Monthly datasets (left), where the covariance matrix is highly unstable due to the small sample size (120 observations), the Sharpe ratio forms a clear hump shape. The regularization successfully filters out statistical noise before eventually overpowering the matrix at excessively high γ values, forcing the portfolio to simply become the Equally Weighted ($1/N$) strategy. In stark contrast, the Daily datasets (right) behave completely differently. Because the daily estimates are built on a massive sample size (2,520 observations), the unconstrained sample covariance matrix is already naturally stable. As a result, adding a penalty generally provides no benefit. For the large datasets like 25 and 100 Size, the Sharpe ratio strictly declines as γ increases, proving that if a covariance matrix is already well-conditioned, introducing an penalty simply adds bias without any necessary variance reduction.

2.3 Interpretation of Portfolio Weights

To definitively answer how the weights compare to the sample Minimum-Variance portfolio, we examine the structural shrinkage of the long and short allocations.

Dataset	Avg Abs L_2	Max Abs L_2	Min Weight L_2	Max Weight L_2	Avg Abs MV	Max Abs MV
10 Ind Daily	0.138	0.506	-0.086	0.506	0.197	0.795
10 Ind Monthly	0.146	0.404	-0.141	0.404	0.217	0.590
25 Size Daily	0.217	0.752	-0.482	0.752	0.218	0.753
25 Size Monthly	0.266	0.741	-0.573	0.716	0.306	0.888
48 Ind Daily	0.038	0.161	-0.062	0.161	0.064	0.466
48 Ind Monthly	0.043	0.176	-0.080	0.176	0.115	0.494
100 Size Daily	0.054	0.379	-0.169	0.379	0.054	0.379
100 Size Monthly	0.073	0.276	-0.215	0.272	0.239	0.969

Table 3: Comparison of portfolio weights between the ℓ_2 -regularized and sample minimum-variance portfolios.

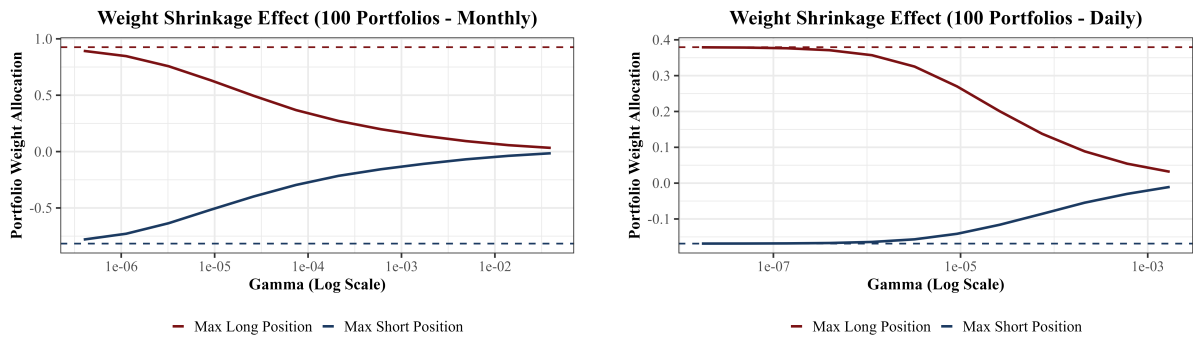


Figure 7: Weight shrinkage effect for the 100-Portfolio dataset in monthly (left) and daily (right) frequencies. As γ increases, extreme leverage is aggressively penalized.

As shown in Table 3 and visually confirmed in Figure 7, the ℓ_2 penalty aggressively forces extreme weights to contract. For the unstable monthly 100-Portfolio dataset (left), the unconstrained Minimum-Variance baseline takes highly leveraged long and short positions approaching $\pm 100\%$ (dashed lines). As the γ penalty increases (moving right along the x-axis), these maximum long and short positions are forcefully shrunk toward zero, eventually converging into a tight, balanced allocation. This proves that the ℓ_2 penalty successfully prevents the optimizer from taking extreme, unrealistic bets based on noisy historical data.

The daily dataset of the same portfolio (right) provides visual proof of our previous performance conclusion. Because the daily estimates rely on 2,520 observations, the unconstrained Minimum-Variance weights (dashed lines) are already tightly bounded between roughly $+40\%$ and -20% . Applying the ℓ_2 penalty here forces the optimizer to shrink an allocation that is already stable. Instead of fixing a broken matrix, the penalty acts as an unnecessary constraint, blindly pushing the portfolio toward equal weighting. We know this damages the portfolio because, as shown previously in Figure 6, the Sharpe ratio for the large daily datasets strictly declines the moment γ increases. This proves that adding bias to an already well-conditioned matrix provides no risk-adjusted benefit.

3 Dynamic γ Selection via Cross-Validation

In Question 2, we sort of cheated by identifying the single best γ using the entire out-of-sample period. In reality, an investor cannot look into the future. The regularization penalty must be chosen dynamically using only the historical data available at each rebalancing step. To make this strategy realistic, we implement a dynamic γ selection process using k -fold cross-validation.

3.1 Cross-Validation Procedure

Within each rolling estimation window, we must evaluate our penalty parameters without peeking into the actual out-of-sample test period. We apply a k -fold cross-validation procedure. Because financial returns are time-series data with inherent autocorrelation and volatility clustering, we cannot randomly shuffle the days or months. Instead, we preserve the chronological order by dividing the data sequentially. The procedure operates as follows:

1. **Grid Setup:** Define a grid of candidate γ values, spaced logarithmically to test a wide range of penalty magnitudes.
2. **Sequential Splitting:** Divide the estimation window into k time blocks (folds). The number of folds is chosen dynamically based on the dataset's sampling frequency to ensure each validation block contains enough observations to calculate a meaningful out-of-sample variance. For the daily datasets (2,520 observations), we use $k = 10$ folds, resulting in sequential 1-year testing blocks. For the monthly datasets (120 observations), using 10 folds would leave only 12 months per block—a sample too small to measure portfolio variance reliably. Therefore, we restrict the monthly data to $k = 5$ folds, resulting in safer, 2-year testing blocks.
3. **Training Phase:** For a given γ , hold out one time block to act as the unseen validation set. Combine the remaining $k - 1$ segments to form the training set. Compute the historical sample covariance matrix from this training data, add the γ penalty to its diagonal, and extract the resulting constrained weights.
4. **Testing Phase:** Apply these computed weights to the actual returns in the time block that was left out. This allows us to simulate how the portfolio would have performed on new, unseen data.
5. **Iteration & Selection:** Repeat steps 3 and 4 until every single segment has served as the validation set exactly once. Calculate the variance of the combined validation returns. The γ that produces the absolute lowest cross-validated variance is declared the winner.

After selecting the optimal γ for that specific rolling window, we compute the final ℓ_2 -regularized portfolio weights using the entire 10-year training window, and evaluate those weights over the actual six-month out-of-sample period. Importantly, we explicitly choose to minimize cross-validated variance rather than maximize the cross-validated Sharpe ratio. This strictly aligns with our Minimum-Variance objective function and ensures we do not rely on estimated mean returns ($\hat{\mu}$), which are notoriously noisy and unstable in portfolio selection.

3.2 Empirical Results

Table 4 reports the final out-of-sample performance when γ is selected dynamically without future information.

Dataset	Ann. Mean	Ann. Vol	CV Sharpe	Fixed γ Sharpe
10 Ind Monthly	12.19%	12.31%	0.991	1.016
10 Ind Daily	12.45%	13.37%	0.931	0.932
25 Size Monthly	16.63%	12.91%	1.288	1.341
25 Size Daily	19.77%	13.09%	1.510	1.510
48 Ind Monthly	11.86%	11.89%	0.997	1.016
48 Ind Daily	11.56%	12.30%	0.940	0.949
100 Size Monthly	17.37%	13.10%	1.326	1.407
100 Size Daily	19.06%	11.02%	1.729	1.764

Table 4: Out-of-sample performance using dynamically selected γ via cross-validation, directly compared against the best fixed- γ baselines established in section 2.

The results show that the cross-validated strategy provides highly robust out-of-sample performance across the board. By dynamically adapting to the noise levels in the historical data, the cross-validation algorithm produces Sharpe ratios that are remarkably close to the best fixed- γ portfolios from question 2. For instance, the highly unstable 100 Size monthly dataset achieves a Sharpe ratio of 1.326. This captures nearly all the upside of the optimal fixed- γ portfolio (1.407). This proves that the algorithm successfully identified the necessary penalty to stabilize the matrix without needing any future information.

3.3 Evolution of Optimal γ

The selected values of γ differ significantly across datasets and frequencies, as detailed in Table 5.

Dataset	Avg γ	Median γ	Min γ	Max γ
10 Ind Monthly	5.74e-04	3.63e-04	1.62e-05	3.99e-03
10 Ind Daily	4.22e-06	1.99e-06	1.99e-08	2.65e-05
25 Size Monthly	2.38e-04	1.71e-04	2.34e-05	1.40e-03
25 Size Daily	1.95e-06	5.64e-07	7.44e-09	1.21e-05
48 Ind Monthly	1.68e-03	8.27e-04	2.78e-04	6.07e-03
48 Ind Daily	1.11e-05	5.66e-06	1.39e-08	5.41e-05
100 Size Monthly	1.21e-03	6.97e-04	1.95e-04	1.17e-02
100 Size Daily	3.81e-06	3.29e-06	1.60e-08	1.58e-05

Table 5: Summary statistics of the dynamically selected γ penalties across all rebalancing windows.

As Table 5 shows, the size of the optimal penalty depends heavily on the data inside the rolling window. Two key patterns stand out:

First, daily data requires a much smaller penalty than monthly data. Because the daily datasets contain 2,520 observations per window, the covariance matrix is already estimated with high accuracy. The algorithm sees this and consistently selects tiny γ values (around 10^{-6}). On the other hand, the monthly datasets only have 120 observations. This forces the algorithm to apply much larger penalties (around 10^{-3}) to fix the noisy matrix.

Second, portfolios with more assets need larger penalties. When the number of assets (N) gets closer to the number of time observations (T), the matrix becomes very unstable. This is obvious in the monthly data, where the large 48- and 100-asset portfolios need the highest average γ values (1.68×10^{-3} and 1.21×10^{-3}) to stop the optimizer from taking extreme long and short positions.

Figure 8 visually demonstrates how this dynamic selection works.

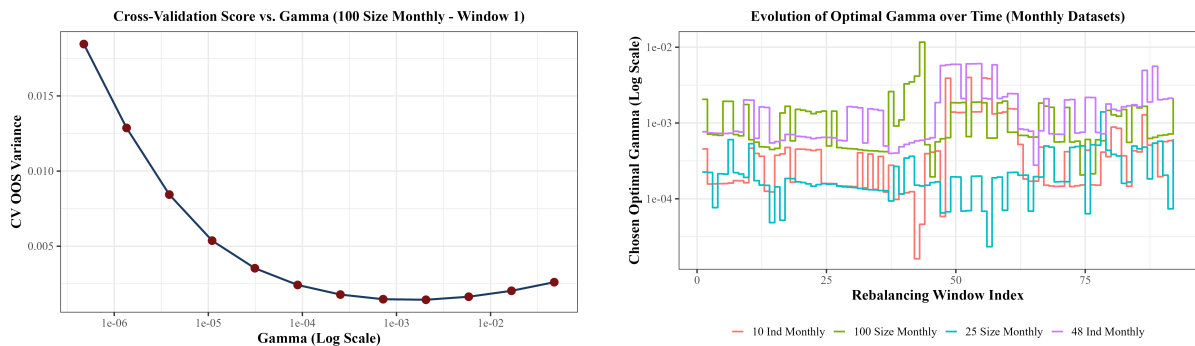


Figure 8: Left: The classic U-Shape cross-validation curve showing the variance minimization for a single window. Right: The evolution of the optimal γ chosen at each rebalancing step across the monthly datasets.

The left plot illustrates why we need to optimize γ . If the penalty is too small, estimation error ruins the portfolio. If it is too big, the penalty ignores useful historical correlation data. The clear U-shape proves that our grid search successfully finds the spot that minimizes out-of-sample variance. The right plot shows how the optimal penalty adapts over time. When rolling windows capture periods of high market stress, historical correlations become unreliable. The algorithm detects this noise and automatically increases γ to protect the portfolio. When the market calms down, it safely lowers the penalty again.

3.4 Evaluation of the Dynamic Strategy

The results confirm that ℓ_2 regularization is a powerful tool for stabilizing portfolio weights, but choosing the magnitude of γ is non-trivial. Cross-validation provides a highly realistic, out-of-sample methodology to balance the bias-variance trade-off dynamically. It successfully prevents catastrophic weight explosions and delivers highly competitive risk-adjusted returns across both asset classes and sampling frequencies, effectively closing the gap between theoretical optimization and practical implementation.

4 Alternative Penalties (Lasso and Elastic Net)

In the previous section, we demonstrated that applying an ℓ_2 (Ridge) penalty successfully stabilizes the sample covariance matrix and prevents extreme portfolio weights. However, while Ridge shrinks weights toward zero, it rarely forces them to exactly zero. This means the portfolio remains exposed to all assets, even highly volatile ones. In this section, we expand our framework to determine whether enforcing strict sparsity, actively filtering out noisy assets by setting their weights to exactly zero, can further enhance out-of-sample risk-adjusted returns.

4.1 Methodology

To test this hypothesis, we introduce two alternative portfolio-weight penalties to reproduce our minimum-variance analysis. The 1-norm-constrained (Lasso) minimum-variance portfolio applies an ℓ_1 penalty, mathematically defined as the sum of the absolute values of the portfolio weights ($\|w\|_1 = \sum_{i=1}^N |w_i|$). By penalizing the absolute size of the weights directly, the optimization algorithm is forced to push the weights of noisy or highly correlated assets to exactly zero to minimize the cost function. This allows Lasso to act as an aggressive feature selector, completely filtering out unstable assets from the portfolio rather than just down-weighting them. It is defined as:

$$\hat{w}_\gamma^{\text{LASSO}} = \arg \min_{w' \mathbf{1} = 1} w' \hat{\Sigma} w + \gamma \|w\|_1$$

The Elastic Net penalty bridges the two approaches we have seen so far by applying a convex combination of the ℓ_1 and ℓ_2 norms. This hybrid formulation retains the aggressive noise-filtering (sparsity) of Lasso while preserving the stabilizing properties of Ridge, which is particularly useful when dealing with highly correlated assets. The Elastic Net-constrained minimum-variance portfolio is defined as:

$$\hat{w}_{\gamma, \alpha}^{\text{EN}} = \arg \min_{w' \mathbf{1} = 1} w' \hat{\Sigma} w + \gamma (\alpha \|w\|_1 + (1 - \alpha) w' w)$$

In our analysis we set $\alpha = 0.5$ to apply an equal weight to both regularization effects.

Because the Lasso penalty uses absolute values, we cannot solve for the weights using standard matrix formulas. Instead, we have to treat the optimization as a linear programming problem and use a numerical solver to find the best weights, always making sure the final weights sum to 1 (fully invested portfolio).

To find the best γ dynamically, we follow the exact same 10-fold cross-validation procedure established in the previous section. Inside each 10-year rolling window, we test our candidate γ values and select the one that yields the lowest average out-of-sample variance across the folds. We explicitly maintain variance minimization as our validation criterion to ensure the cross-validation remains strictly aligned with our minimum-variance objective function. By repeating this process recursively, the algorithms automatically adapt their penalties to shifting market conditions.

4.2 Empirical Results

Table 6 reports the out-of-sample annualized mean, volatility, and Sharpe Ratio obtained with both the Lasso and Elastic Net optimizations across all datasets and sampling frequencies.

Dataset	Frequency	Methodology	OOS Mean	OOS Vol	OOS SR
25 Size-B/M	Monthly	LASSO	16.68%	13.27%	1.257
25 Size-B/M	Monthly	Elastic Net	16.59%	12.97%	1.279
25 Size-B/M	Daily	LASSO	19.85%	13.13%	1.512
25 Size-B/M	Daily	Elastic Net	19.84%	13.12%	1.513
100 Size-B/M	Monthly	LASSO	18.27%	12.97%	1.408
100 Size-B/M	Monthly	Elastic Net	18.41%	13.15%	1.400
100 Size-B/M	Daily	LASSO	19.44%	11.09%	1.753
100 Size-B/M	Daily	Elastic Net	19.43%	11.09%	1.752
10 Industry	Monthly	LASSO	12.25%	12.37%	0.990
10 Industry	Monthly	Elastic Net	12.17%	12.36%	0.985
10 Industry	Daily	LASSO	12.26%	13.39%	0.915
10 Industry	Daily	Elastic Net	12.30%	13.39%	0.919
48 Industry	Monthly	LASSO	11.00%	12.15%	0.905
48 Industry	Monthly	Elastic Net	11.02%	12.41%	0.888
48 Industry	Daily	LASSO	11.35%	12.34%	0.920
48 Industry	Daily	Elastic Net	11.38%	12.34%	0.922

Table 6: Out-of-sample performance results of LASSO and Elastic Net minimum-variance portfolios.

While both methodologies provide highly robust performance across the board, their true advantage emerges in high-dimensional, noisy environments. For instance, in the highly unstable 100 Size Monthly dataset, the Lasso and Elastic Net strategies drive the Sharpe ratio up to 1.408 and 1.400 respectively, significantly outperforming the standard ℓ_2 baseline (1.117) established in section 3. Crucially, they achieve this not by drastically increasing returns, but by severely crushing portfolio volatility.

To understand the mechanics driving this outperformance, we visualize both the cumulative wealth generation and the portfolio sparsity for this specific dataset in Figure 9.

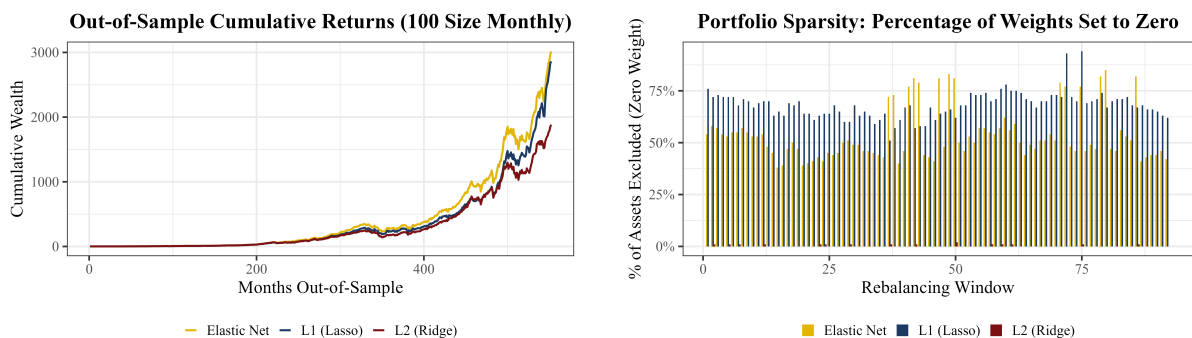


Figure 9: Left: Out-of-sample cumulative returns for the 100 Size monthly dataset. Right: The sparsity effect, measuring the percentage of assets assigned a weight of exactly zero at each rebalancing step.

The right panel reveals exactly why the new penalties succeed: sparsity. The ℓ_2 penalty (which typically shows near 0% sparsity) keeps almost all 100 assets in the port-

folio, merely shrinking their allocations. In contrast, the Lasso and Elastic Net algorithms automatically exclude a massive percentage of the available assets at every single rebalancing window. By setting the weights of noisy assets to zero, Lasso acts as a built-in noise filter. It prevents the optimizer from taking positions in assets whose historical correlations are too unstable to trust. The left panel proves that this sparse, highly filtered portfolio yields a smoother, less volatile equity curve over time.

Despite its strong out-of-sample performance, the strict ℓ_1 penalty has notable theoretical limitations. In the presence of highly correlated assets, Lasso tends to arbitrarily select one asset while forcing the others to zero, sacrificing potential diversification benefits. Furthermore, the aggressive nature of this feature selection can cause the chosen assets to shift drastically between rebalancing windows, potentially leading to extreme portfolio turnover and excessive transaction costs in a live trading environment. This highlights the exact theoretical value of the Elastic Net penalty: by reintroducing the ℓ_2 norm alongside the absolute value constraint, it stabilizes the selection process, allowing the algorithm to retain groups of correlated assets while still benefiting from Lasso's overarching noise filtration.

5 PCA-Based Covariance Estimation

Principal Component Analysis (PCA) is used to reduce the dimensionality of the covariance estimation problem. This is consistent with the previous sections, where we showed that portfolio optimization is very sensitive to estimation error, especially when the number of assets is large.

The idea is to reconstruct the covariance matrix using only the first K principal components. These components capture the main sources of co-movement between assets. By keeping only the most important components, PCA filters part of the noise contained in the sample covariance matrix and makes the matrix inversion more stable. This is useful for both the Mean-Variance portfolio and the Global Minimum-Variance portfolio (GMV).

5.1 Out-of-Sample Performance as a Function of K

Figures 10–13 show the out-of-sample performance of pure PCA-based portfolios for different values of K . The objective is to understand how the number of retained principal components affects performance.

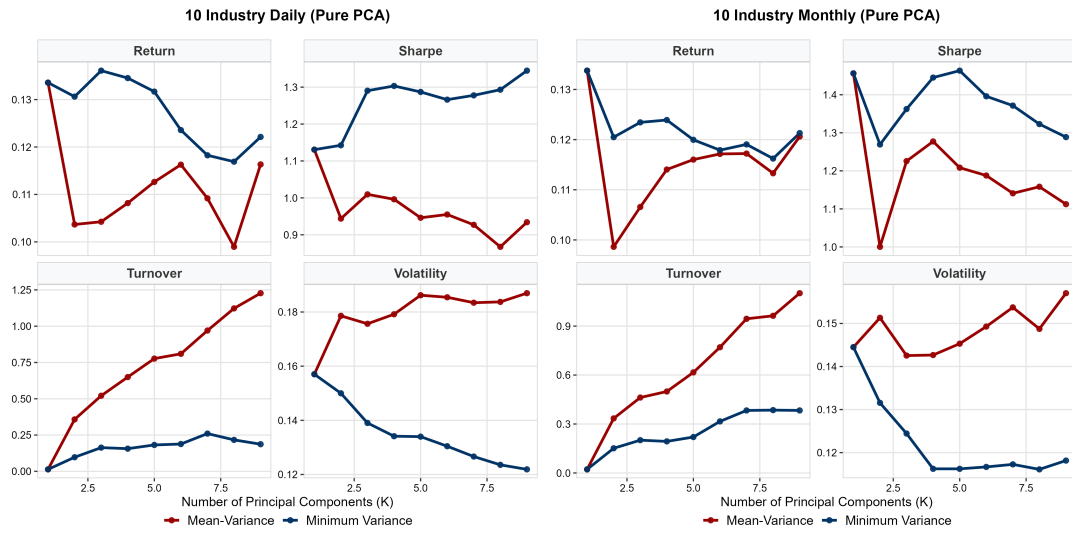


Figure 10: Out-of-sample performance of pure PCA strategies for the 10 Industry dataset.

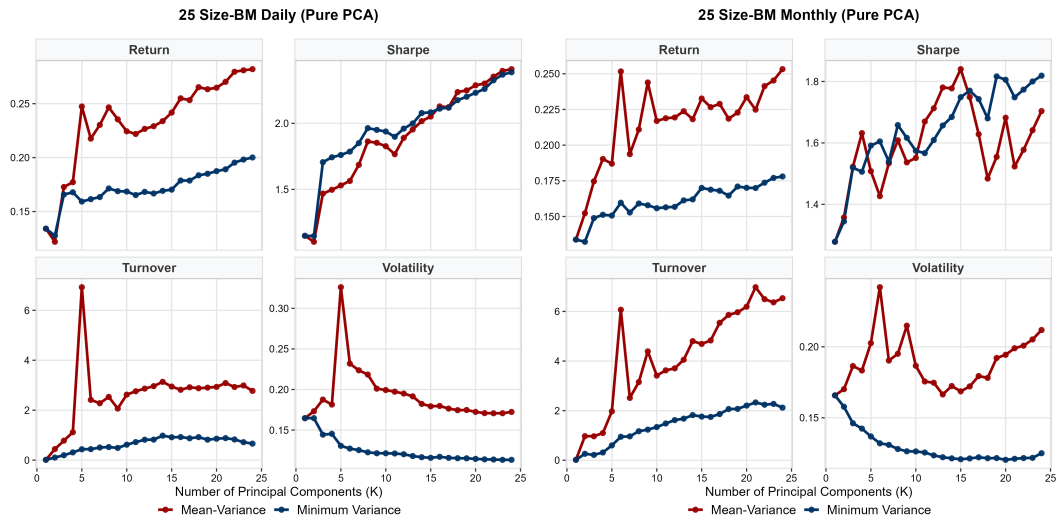


Figure 11: Out-of-sample performance of pure PCA strategies for the 25 Size-BM dataset.

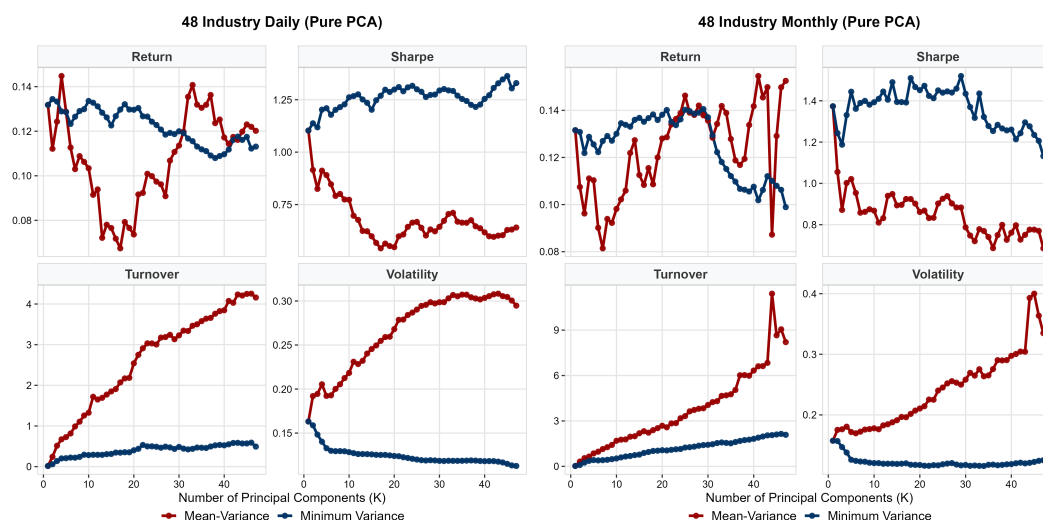


Figure 12: Out-of-sample performance of pure PCA strategies for the 48 Industry dataset.

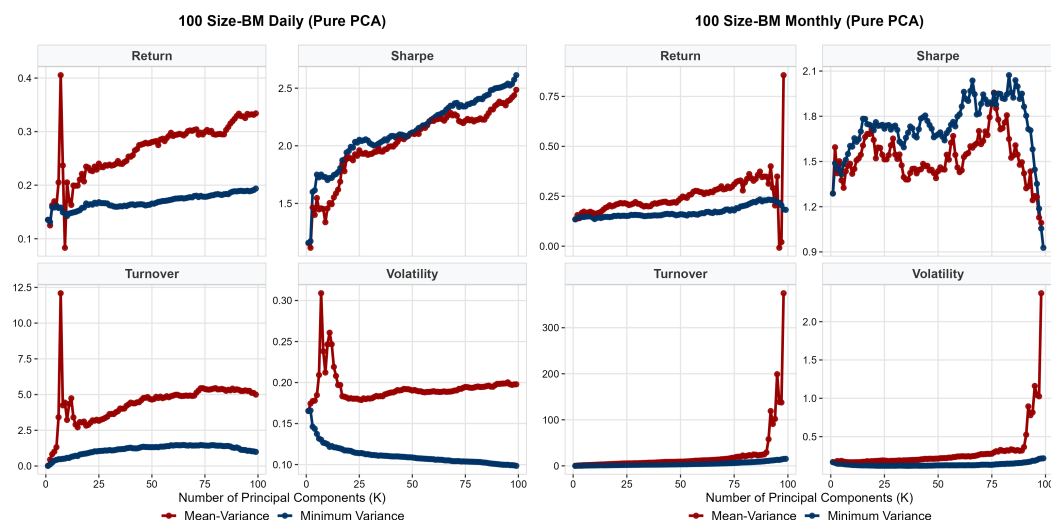


Figure 13: Out-of-sample performance of pure PCA strategies for the 100 Size-BM dataset.

The results show that performance depends strongly on the choice of K . If too few components are kept, the portfolio ignores useful information from the covariance matrix. If too many components are included, noisy components enter the estimation again and can reduce out-of-sample performance.

The Mean-Variance strategy can generate competitive Sharpe ratios in some datasets, but it is less stable because it still depends on estimated expected returns. As discussed in Section 1, expected returns are difficult to estimate precisely, and errors in the mean vector can strongly affect portfolio weights.

By contrast, the GMV strategy is generally more stable because it only relies on the covariance matrix. PCA is therefore especially useful for GMV, because it reduces noise in the covariance structure without requiring the estimation of expected returns.

5.2 Optimal Factor K Selection

To choose the number of principal components, we use the cumulative explained variance. We select the smallest value of K such that the first K components explain at least 90% of the total variance:

$$\text{Explained Variance}(K) = \frac{\sum_{i=1}^K \lambda_i}{\sum_{i=1}^N \lambda_i},$$

where λ_i are the eigenvalues sorted in decreasing order.

This rule provides a simple compromise between information retention and noise reduction. It keeps the main sources of co-movement between assets while excluding components that are more likely to capture estimation noise.

Dataset	Strategy	Return	Vol.	Sharpe	K^*	Exp. Var.
<i>Monthly Datasets</i>						
10IND Monthly	MeanVar	0.1160	0.1453	1.2083	5	0.9191
	GMV	0.1199	0.1162	1.4633		
48IND Monthly	MeanVar	0.1281	0.2106	0.8615	20	0.9009
	GMV	0.1381	0.1174	1.4517		
FF25 Monthly	MeanVar	0.1746	0.1865	1.5225	3	0.9329
	GMV	0.1489	0.1461	1.5193		
FF100 Monthly	MeanVar	0.1994	0.1763	1.6827	18	0.9024
	GMV	0.1474	0.1232	1.7440		
<i>Daily Datasets</i>						
10IND Daily	MeanVar	0.1126	0.1862	0.9461	5	0.9213
	GMV	0.1317	0.1340	1.2870		
48IND Daily	MeanVar	0.1009	0.2841	0.6438	23	0.9002
	GMV	0.1266	0.1217	1.3088		
FF25 Daily	MeanVar	0.1728	0.1875	1.4679	3	0.9227
	GMV	0.1658	0.1443	1.7059		
FF100 Daily	MeanVar	0.2384	0.1805	1.9279	30	0.9020
	GMV	0.1625	0.1121	2.0216		

Table 7: Out-of-sample performance and optimal K^* configuration.

Table 7 confirms that the GMV strategy generally gives stronger and more stable risk-adjusted performance than the Mean-Variance strategy. This is especially visible in larger datasets, such as the 48 Industry and 100 Size-BM portfolios. The reason is that GMV avoids the estimation of expected returns and uses PCA mainly to stabilize the covariance matrix.

The Mean-Variance strategy performs better when the factor space is relatively small, because fewer components reduce the amount of noise entering the optimization. However, when more components are included, the strategy becomes more exposed to noisy mean estimates. Overall, the 90% threshold provides a reasonable balance between keeping relevant information and reducing estimation noise.

5.3 Improving Covariance Estimation with Stein's Shrinkage

A limitation of standard PCA is that the sample eigenvalues are themselves estimated with error. In finite samples, large eigenvalues can be overestimated and small eigenvalues can be underestimated. This may lead to an ill-conditioned covariance matrix and unstable portfolio weights.

To reduce this problem, we apply Stein's shrinkage estimator to the eigenvalues. The regularized eigenvalues are given by:

$$\tilde{\lambda}_i = \frac{T\lambda_i}{\alpha_i}, \quad \text{where} \quad \alpha_i = T - N + 1 + 2\lambda_i \sum_{j \neq i} \frac{1}{\lambda_i - \lambda_j}.$$

In practice, this formula can produce eigenvalues that are not ordered correctly or that are negative. To solve this issue, we apply an isotonic regression correction using the `iso_cov()` function in R. This correction enforces non-increasing and non-negative eigenvalues, which makes the covariance matrix more stable.

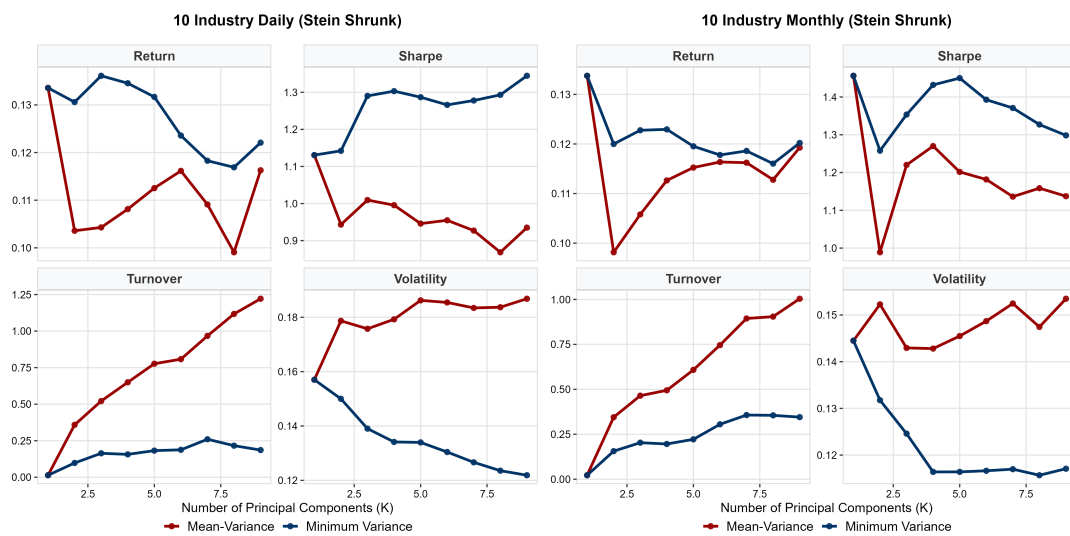


Figure 14: Out-of-sample performance of Stein-shrunk PCA strategies for the 10 Industry dataset.

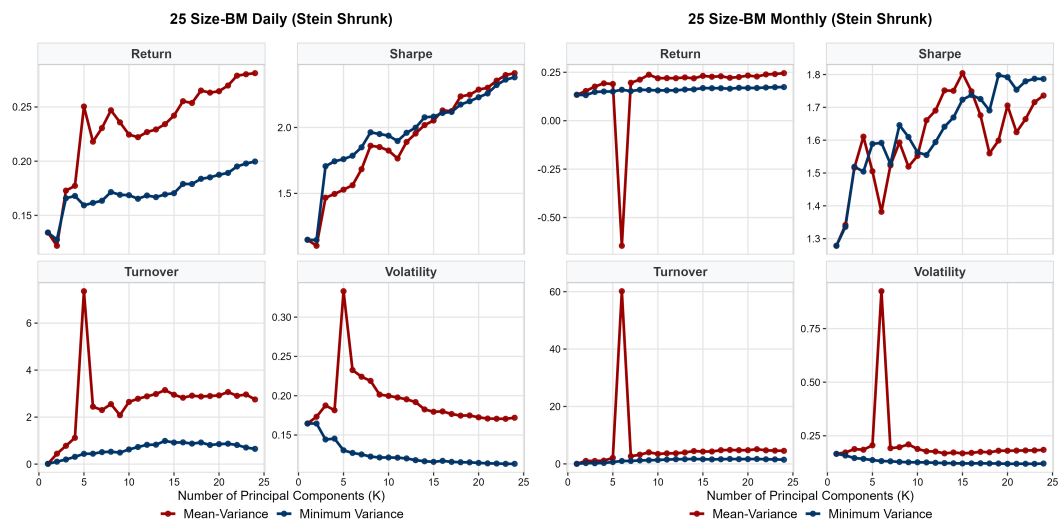


Figure 15: Out-of-sample performance of Stein-shrunk PCA strategies for the 25 Size-BM dataset.

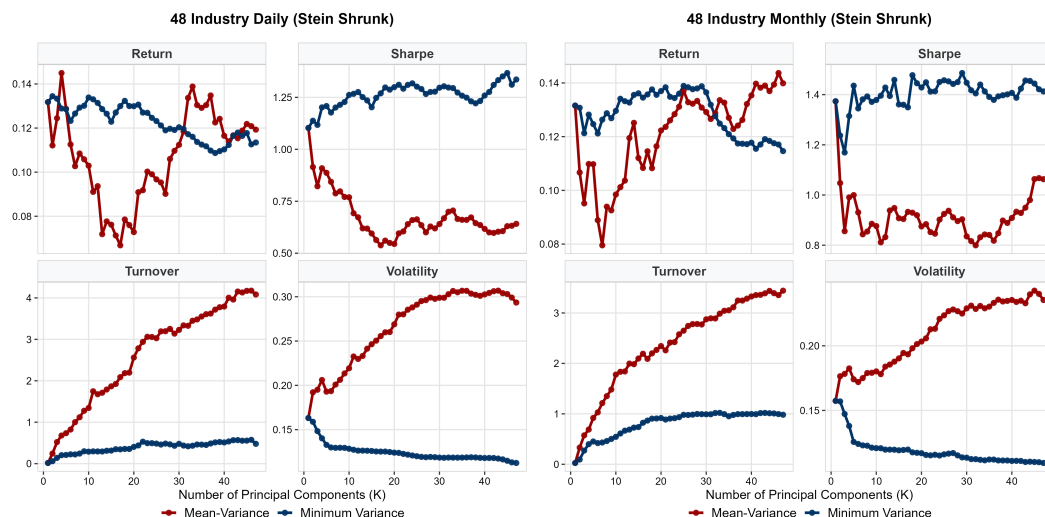


Figure 16: Out-of-sample performance of Stein-shrunk PCA strategies for the 48 Industry dataset.

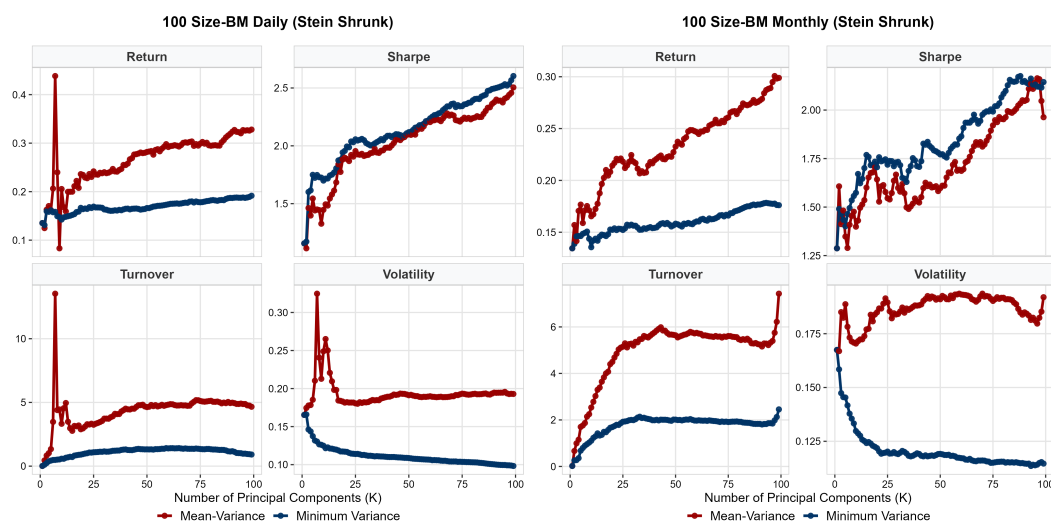


Figure 17: Out-of-sample performance of Stein-shrunk PCA strategies for the 100 Size-BM dataset.

The figures show that Stein shrinkage mainly reduces the most extreme effects caused by noisy eigenvalues. The improvement is not a complete change in performance, but rather a stabilization of the covariance estimate. This is particularly useful for the GMV strategy, because GMV directly depends on the inversion of the covariance matrix.

For the Mean-Variance strategy, the improvement is more limited. Even if the covariance matrix is more stable, the strategy still relies on the estimated mean vector. Therefore, part of the instability remains due to expected return estimation error.

Dataset	Strategy	Return	Vol.	Sharpe	K^*	Exp. Var.
<i>Monthly Datasets</i>						
10IND Monthly	MeanVar	0.1152	0.1455	1.2016	5	0.9191
	GMV	0.1195	0.1164	1.4500		
48IND Monthly	MeanVar	0.1223	0.2033	0.8747	20	0.9009
	GMV	0.1369	0.1169	1.4284		
FF25 Monthly	MeanVar	0.1763	0.1880	1.5163	3	0.9329
	GMV	0.1484	0.1462	1.5190		
FF100 Monthly	MeanVar	0.2047	0.1807	1.6792	18	0.9024
	GMV	0.1481	0.1229	1.7179		
<i>Daily Datasets</i>						
10IND Daily	MeanVar	0.1125	0.1862	0.9460	5	0.9213
	GMV	0.1317	0.1340	1.2868		
48IND Daily	MeanVar	0.1003	0.2854	0.6401	23	0.9002
	GMV	0.1269	0.1218	1.3102		
FF25 Daily	MeanVar	0.1728	0.1876	1.4675	3	0.9227
	GMV	0.1658	0.1443	1.7063		
FF100 Daily	MeanVar	0.2401	0.1818	1.9190	30	0.9020
	GMV	0.1633	0.1122	2.0241		

Table 8: Out-of-sample performance of Stein-isotonized PCA portfolios.

As shown in Table 8, the out-of-sample results under the 90% threshold remain close to the pure PCA results in Table 7. This is expected because the same K^* rule is used in both cases. However, Stein shrinkage slightly stabilizes the eigenvalue spectrum and can make the covariance matrix better conditioned.

The results suggest that Stein shrinkage is more useful for the GMV strategy than for the Mean-Variance strategy. For example, in the FF100 Daily dataset, the GMV Sharpe ratio remains very strong after Stein shrinkage. This confirms that stabilizing the covariance matrix is helpful, especially in high-dimensional settings.

The figures also suggest that using a higher explained-variance threshold, such as 95%, could be tested in future work. Since Stein shrinkage makes additional components less noisy, using more components may become less dangerous than under pure PCA. However, this should be treated as an empirical extension rather than a guaranteed improvement.

5.4 Conclusion

Overall, PCA is useful because it reduces the dimensionality of the covariance estimation problem. It filters part of the noise and helps stabilize portfolio optimization. The choice of K is important: too few components create too much bias, while too many components can reintroduce estimation noise.

Stein's shrinkage provides an additional improvement by stabilizing the eigenvalues used in the covariance matrix. The results are consistent with the main message of the report: robust estimation methods are essential to obtain reliable out-of-sample portfolio performance.

6 Higher Moments and IC-Variance-Parity

Up to this point, our risk analysis has relied exclusively on variance. This reliance implicitly assumed that financial returns follow a symmetric, Gaussian distribution, but this assumption systematically fails in practice. Real-world asset returns consistently exhibit asymmetry (skewness) and fat tails (excess kurtosis). Because the standard Sample Minimum-Variance (MV) portfolio optimizes only the second moment, it remains completely blind to these higher-order dynamics, leaving it highly susceptible to extreme downside risks and severe estimation errors during market turbulence. To address these structural flaws and actively manage crash risk, we implement the Independent Component Analysis (ICA) based variance-parity portfolio.

6.1 Methodology

The primary limitation of Principal Component Analysis (PCA) is that it only ensures that factors are uncorrelated. By definition, PCA identifies a rotation of the data that minimizes reconstruction error and ensures the resulting components have zero linear correlation. While this effectively reduces dimensionality, uncorrelation is a weak condition; it only eliminates linear relationships and remains blind to higher-order non-linear dependencies between factors.

In contrast, the Independent Component Analysis (ICA) approach seeks to extract factors that are statistically independent. Independence is a much stronger condition: it implies that knowing the realization of one factor provides absolutely no information about another. We utilize the FastICA algorithm (Hyvärinen, 1999) to rotate the principal components into a set of independent components (ICs). This algorithm maximizes the negentropy of the factors to identify non-Gaussian market drivers. Specifically, we employ the $G_1(x) = \frac{1}{\alpha} \ln \cosh(\alpha x)$ contrast function, which serves as a robust, general-purpose criterion for isolating these independent signals. Negentropy serves as a proxy for independence because, according to the Central Limit Theorem, the distribution of a mixture of independent variables tends toward a Gaussian distribution. By maximizing non-Gaussianity, we perform the inverse operation: we recover the original, isolated structural sources (ICs) that drive the market.

Following the framework proposed by Lassance et al. (2022), we construct the IC-Variance-Parity (ICVP) portfolio to ensure that risk is spread evenly across the market's statistically independent drivers. We compute the portfolio weights using the unmixing matrix $A = K \times W$, where K denotes the standardization matrix (which scales the principal components to unit variance) and W is the rotation matrix derived from the FastICA algorithm. By satisfying the variance-parity condition, the ICVP portfolio diversifies risk across true economic components rather than individual, correlated assets.

To evaluate the portfolio's resilience against tail events, we compute the Modified Value-at-Risk (mVaR) at a 95% confidence level. We employ the Cornish-Fisher expansion to adjust the standard VaR calculation, explicitly incorporating the portfolio's realized skewness (ζ_P) and excess kurtosis (κ_P):

$$\text{mVaR}_\alpha(P) = -\mu_P - \sigma_P \left(z_\alpha + \frac{1}{6}(z_\alpha^2 - 1)\zeta_P + \frac{1}{24}(z_\alpha^3 - 3z_\alpha)\kappa_P - \frac{1}{36}(2z_\alpha^3 - 5z_\alpha)\zeta_P^2 \right)$$

6.2 Empirical Results Analysis

Table 9 presents the out-of-sample performance, higher moments, and mVaR for the ICVP portfolios across different values of K , compared against the Sample Minimum-Variance baseline.

Strategy	Ann. Mean	Ann. Vol	Skewness	Exc. Kurtosis	Mod. VaR (95%)
IC-Variance-Parity $K = 1$	0.1345	0.1843	-0.7010	2.5543	-0.0835
IC-Variance-Parity $K = 5$	0.1753	0.1819	-0.1330	1.4864	-0.0720
IC-Variance-Parity $K = 10$	0.1382	0.1773	-0.3207	2.1733	-0.0749
IC-Variance-Parity $K = 25$	0.1617	0.1687	-0.3739	2.3500	-0.0693
IC-Variance-Parity $K = 50$	0.1627	0.1611	-0.2922	1.6338	-0.0651
IC-Variance-Parity $K = 75$	0.1670	0.1678	-0.5761	2.1339	-0.0712
IC-Variance-Parity $K = 100$	0.1826	0.2603	-0.0883	-0.0401	-0.1102
Sample MinVar	0.1814	0.2557	-0.1400	0.0839	-0.1090

Table 9: Out-of-sample performance, higher moments, and Modified VaR (95%) for ICVP strategies compared to the Sample Minimum-Variance portfolio.

The results demonstrate a compelling improvement in tail risk management. The Sample Minimum-Variance portfolio exhibits high sensitivity to estimation noise when applied to a large universe of assets ($N = 100$), which severely inflates its out-of-sample volatility to 25.57%. In contrast, the ICVP strategy ($K = 50$) succeeds in filtering out the estimation noise that plagues the standard MV approach, crushing portfolio volatility down to 16.11%. By isolating independent factors, the ICVP portfolio significantly reduces the 95% Modified VaR from a severe -10.90% (Sample MV) to a much safer -6.51%, effectively lowering the catastrophic tail risk for the investor.

As we increase the number of independent components K , we observe an evolution in the portfolio's moment profile. At low values of K (e.g., $K = 1$), the portfolio is overly concentrated on the primary market factors, leading to higher negative skewness (-0.70) and elevated kurtosis (2.55). As K increases toward the optimal range, the FastICA algorithm effectively diversifies the portfolio across a richer set of independent structural drivers. This structural diversification results in a lower excess kurtosis and a more favorable skewness profile, which directly translates into a structurally safer Modified VaR. However, when K approaches N ($K = 100$), the algorithm begins to capture pure estimation noise, causing the performance metrics to revert toward the unstable, high-volatility characteristics of the standard sample-based estimation (reaching 26.03% volatility and a -11.02% mVaR). Thus, the ICVP methodology confirms that managing independent components, rather than simple variances, is a superior approach for tail-risk-aware portfolio construction.

7 Additional Portfolio Strategy: ICMV

7.1 Methodology and Objective

For this additional strategy, we implement the ICMV shrinkage portfolio introduced by Lassance et al. (2022). This strategy serves as a hybrid framework, combining the classical Global Minimum-Variance (GMV) portfolio with a risk-parity portfolio based on Independent Components (ICs).

The portfolio is defined as:

$$\mathbf{w}_{ICMV} = (1 - \delta)\mathbf{w}_{MV} + \delta\mathbf{w}_{IC},$$

where $\mathbf{w}_{MV}$ denotes the sample minimum-variance portfolio and $\mathbf{w}_{IC}$ represents the IC-variance-parity portfolio. The parameter $\delta \in [0, 1]$ functions as a shrinkage intensity, controlling the allocation balance between the two strategies.

When $\delta = 0$, the strategy collapses to the standard GMV portfolio; when $\delta = 1$, it becomes the pure IC-variance-parity portfolio. Intermediate values allow the investor to capture the low-volatility stability of the GMV approach while simultaneously harvesting the non-Gaussian diversification benefits inherent in the IC framework. Our objective is to reduce tail risk in non-normal return distributions, specifically by addressing negative skewness and excess kurtosis, where standard variance-based models often prove inadequate.

7.2 Calibration of δ

The optimal shrinkage intensity δ^* is calibrated dynamically within each rolling estimation window. We conduct a grid search over $\delta \in [0, 1]$ and select the value that maximizes the modified Sharpe ratio:

$$\delta^* = \arg \max_{\delta \in [0, 1]} \frac{\mu_P(\delta) - R_f}{MVaR_\alpha(P, \delta)}.$$

To compute the modified Value-at-Risk, we utilize the same Cornish-Fisher expansion framework employed in section 6, which effectively accounts for the portfolio's realized asymmetry and fat tails:

$$MVaR_\alpha(P) = -\mu_P - \sigma_P \left(z_\alpha + \frac{1}{6}(z_\alpha^2 - 1)\zeta_P + \frac{1}{24}(z_\alpha^3 - 3z_\alpha)\kappa_P - \frac{1}{36}(2z_\alpha^3 - 5z_\alpha)\zeta_P^2 \right)$$

where z_α is the Gaussian quantile, ζ_P is the portfolio skewness, and κ_P is the excess kurtosis ($\alpha = 1\%$). This calibration ensures the strategy is directly tailored to the investor's specific risk sensitivity, prioritizing comprehensive tail-risk mitigation, just as we explored during our IC-Variance-Parity analysis in section 6, rather than relying solely on classical Sharpe maximization.

After selecting the optimal δ^* , the corresponding ICMV portfolio is held out-of-sample over the forward six-month testing block, utilizing the identical rolling-window methodology established throughout our empirical framework.

7.3 ICMV Deltas and Allocations

Table 10 summarizes the average calibrated δ values and the corresponding annualized risk-return profiles.

The moderate values of δ confirm that the strategy maintains its minimum-variance anchor while selectively tilting toward IC-based diversification. Interestingly, we observe that the optimal δ is highly dependent on the empirical structure and observation frequency of the dataset; daily datasets generally command higher δ values, likely because the increased density of observations allows for more robust estimation of higher-order moments.

Dataset	Frequency	K	Avg. δ	GMV σ_p	GMV μ_p	ICMV σ_p	ICMV μ_p
10 Industry	Daily	5	0.3929	0.1320	0.1156	0.1434	0.1135
25 Portfolios	Daily	10	0.1745	0.1293	0.1601	0.1320	0.1602
48 Industry	Daily	20	0.2815	0.1209	0.1057	0.1312	0.1067
100 Portfolios	Daily	25	0.1315	0.1069	0.1748	0.1085	0.1746
10 Industry	Monthly	5	0.1995	0.1233	0.1161	0.1248	0.1149
25 Portfolios	Monthly	10	0.1071	0.1232	0.1497	0.1236	0.1496
48 Industry	Monthly	20	0.2277	0.1103	0.1062	0.1135	0.1115
100 Portfolios	Monthly	25	0.1582	0.1092	0.1516	0.1110	0.1518

Table 10: ICMV allocation parameters. μ_p and σ_p are annualized.

7.4 Out-of-sample performance

Table 11 shows our out-of-sample results. The ICMV strategy is essentially a trade-off: it doesn't aim to beat the standard GMV portfolio on the Sharpe ratio every time, but it offers a big advantage in daily-frequency data where tail risks are much harder to ignore.

In smaller datasets (like 10-Industry or 25-Portfolio), GMV often holds a higher Sharpe ratio, as seen in the 10-Industry Daily set where it falls from 0.913 to 0.686. However, in larger datasets, ICMV often comes out on top; for example, the 48-Industry Monthly Sharpe ratio improves from 0.701 to 0.813.

The real strength of ICMV shows up in the higher moments. Since daily returns are prone to fat tails, ICMV effectively brings down excess kurtosis. In the 10-Industry Daily set, kurtosis drops from 19.75 to 12.17, and in the 48-Industry Daily set, it falls from 20.41 to 14.02. While a pure IC strategy can reduce kurtosis even more, it usually comes with too much volatility. ICMV effectively provides the best of both worlds: it keeps the stability of GMV while using just enough IC-based diversification to dampen extreme market moves.

The modified Value-at-Risk results are mixed. In some datasets, ICMV improves the modified VaR compared with GMV, such as the 10 Industry Daily, 25 Portfolio Daily, 48 Industry Daily and 100 Portfolio Monthly datasets. In other datasets, the improvement is limited or absent. This confirms that the benefit of ICMV depends on the dataset and on the return frequency.

Dataset	Strategy	Mean	Vol	Sharpe	Skew.	Kurt.	MVaR
10 Ind Monthly	GMV	0.1217	0.1240	0.9816	-0.3203	0.4785	0.2916
	ICMV	0.1159	0.1288	0.8999	-0.3948	0.6533	0.3156
	IC	0.1056	0.1588	0.6650	-0.3956	0.9070	0.4095
10 Ind Daily	GMV	0.1224	0.1340	0.9130	-0.2313	19.7507	0.9430
	ICMV	0.1195	0.1742	0.6861	-0.0794	11.2927	0.8672
	IC	0.1273	0.2309	0.5514	-0.1290	8.2530	0.9951
25 Port Monthly	GMV	0.1791	0.1355	1.3219	-0.2646	0.8085	0.3118
	ICMV	0.1788	0.1375	1.3008	-0.1514	1.0032	0.3145
	IC	0.1507	0.1713	0.8801	-0.1748	1.3651	0.4296
25 Port Daily	GMV	0.1987	0.1316	1.5099	-0.2878	24.2740	1.0640
	ICMV	0.1989	0.1362	1.4601	-0.2729	21.0910	0.9997
	IC	0.1719	0.1883	0.9132	-0.0946	8.2661	0.8034
48 Ind Monthly	GMV	0.0997	0.1423	0.7006	-0.1614	0.8743	0.3469
	ICMV	0.1140	0.1402	0.8130	-0.2701	1.0709	0.3524
	IC	0.1487	0.1525	0.9751	-0.1722	1.8358	0.3950
48 Ind Daily	GMV	0.1111	0.1241	0.8953	-0.5847	20.4143	0.9112
	ICMV	0.1129	0.1352	0.8351	-0.4807	14.0512	0.7876
	IC	0.1281	0.1946	0.6580	-0.1459	3.7944	0.6367
100 Port Monthly	GMV	0.1814	0.2557	0.7094	-0.1400	0.0839	0.5720
	ICMV	0.1972	0.2371	0.8320	-0.1786	0.4745	0.5492
	IC	0.1636	0.1687	0.9698	-0.3626	2.3114	0.4730
100 Port Daily	GMV	0.1951	0.1106	1.7645	-1.0056	16.2700	0.7052
	ICMV	0.2017	0.1131	1.7831	-0.9704	15.8186	0.7092
	IC	0.1755	0.1865	0.9411	-0.2386	13.7459	1.0506

Table 11: Out-of-sample performance metrics.

7.5 Conclusion

Overall, the ICMV strategy is useful because it offers a compromise between classical minimum-variance optimization and higher-moment diversification. It does not always improve the classical Sharpe ratio, but it can improve tail-risk measures and reduce excess kurtosis in several datasets, especially with daily returns.

The main conclusion is therefore that ICMV is not a uniformly dominant strategy, but it is relevant for investors who care about more than mean and volatility. It is particularly interesting when the objective is to control non-Gaussian risks such as skewness, excess kurtosis and modified Value-at-Risk.

8 Main Conclusions

Based on the analysis conducted in the previous sections, we draw three main conclusions.

8.1 Robustness is more important than in-sample optimization

The first main conclusion is that good in-sample performance does not guarantee good out-of-sample performance. In Section 1, the sample mean-variance portfolio often performs very well in-sample because it uses the full historical sample and strongly exploits the estimated mean returns. However, its out-of-sample performance is much less stable, especially for large datasets such as the 100 size-and-book-to-market portfolios.

This result is mainly due to estimation error. The mean-variance portfolio is very sensitive to the estimated mean vector, which is difficult to estimate precisely. Small errors in expected returns can lead to very large long and short positions. This explains why simpler strategies, such as the equally weighted portfolio or the minimum-variance portfolio, can sometimes perform better out of sample.

This idea is also confirmed in Section 7. The ICMV strategy does not dominate the classical GMV portfolio in all datasets, even though it is more sophisticated. For example, ICMV improves the Sharpe ratio in some larger datasets, such as the 48 Industry Monthly and 100 Portfolio Monthly datasets, but GMV remains better in other cases. This shows again that a more advanced strategy is not automatically better out of sample.

Therefore, one important lesson is that the most optimized or complex strategy is not necessarily the best strategy in practice. A portfolio strategy must be robust to estimation error, not only optimal on past data.

8.2 Regularization and shrinkage improve portfolio stability

The second conclusion is that constraints, penalties and shrinkage methods are useful to improve robustness. In Sections 2, 3 and 4, we tested several regularized minimum-variance portfolios, including the ℓ_2 penalty, LASSO and Elastic Net.

The ℓ_2 penalty reduces the magnitude of extreme weights by penalizing large values of $w'w$. This makes the portfolio less sensitive to noise in the estimated covariance matrix. The results show that regularized portfolios often produce more stable weights and competitive out-of-sample Sharpe ratios.

The choice of the regularization parameter is also important. If the penalty is too weak, the portfolio remains close to the unconstrained minimum-variance portfolio and may still be unstable. If the penalty is too strong, the portfolio becomes too close to the equally weighted portfolio and may ignore useful information from the covariance matrix. Section 3 shows that cross-validation provides a realistic way to choose this parameter using only past data, even if it does not always select the value that gives the best realized Sharpe ratio ex post.

Section 7 gives another example of the same idea through the ICMV shrinkage portfolio. ICMV combines the GMV portfolio with an independent-component portfolio using a shrinkage intensity δ . This creates a compromise between low volatility and higher-moment diversification. The results show that ICMV can improve performance in some datasets, but the benefit depends on the data and on the value of δ .

Overall, regularization and shrinkage are useful because they introduce some bias, but they can strongly reduce estimation variance. This bias-variance trade-off is central in portfolio optimization.

8.3 The estimation method and the risk measure matter

The third conclusion is that portfolio performance depends strongly on how we estimate the inputs and how we define risk. In Section 5, PCA was used to reduce the dimension of the covariance matrix. By keeping only the first principal components, we reduce the number of parameters to estimate and filter part of the noise. This can improve the stability of the portfolio, especially in large-dimensional datasets.

However, the choice of the number of components K is important. If K is too small, the portfolio ignores too much information. If K is too large, noisy components are included again. The results with Stein's shrinkage show that correcting the eigenvalues can make the covariance matrix more stable and reduce the risk of using noisy sample eigenvalues.

Sections 6 and 7 also show that variance is not the only relevant risk measure. Financial returns are not normally distributed and often have negative skewness and excess kurtosis. The ICA-based approach helps to study higher moments by diversifying across independent components instead of only relying on correlations. This is useful because a portfolio with low volatility can still be exposed to tail risk.

In Section 7, the ICMV results confirm this point. ICMV does not always improve the classical Sharpe ratio, but it often improves some tail-risk indicators. In particular, for several daily datasets, ICMV reduces excess kurtosis compared with GMV. This means that evaluating a strategy only with mean, volatility and Sharpe ratio can be incomplete. Skewness, excess kurtosis and modified Value-at-Risk should also be considered when the investor cares about downside risk and extreme returns.

8.4 Overall conclusion

The general conclusion of the project is that out-of-sample portfolio performance depends mainly on robustness. A good portfolio strategy should avoid overfitting, control extreme weights, and use estimation methods adapted to the number of assets and the sample size. In practice, simple and regularized strategies often perform better than highly optimized strategies because they are less sensitive to estimation error.

The project also shows that risk cannot be summarized only by volatility. Higher moments and tail-risk measures are important, especially when returns are non-normal. The best approach is therefore not necessarily the most complex one, but the one that finds a good balance between using information from the data, controlling estimation risk, and accounting for the full distribution of portfolio returns.

9 Best Practices for Out-of-Sample Portfolio Performance

Based on the course, the academic papers used in this project, and our empirical results, we can identify several best practices to improve out-of-sample portfolio performance.

9.1 Focus on robustness rather than in-sample fit

The main objective of portfolio construction should not be to obtain the best in-sample performance. Instead, the objective should be to build a portfolio that remains stable

when applied out of sample. This is especially important in a big-data setting, where the number of assets or inputs can be large relative to the number of observations.

A portfolio that performs extremely well in sample may simply be overfitting historical data. This was one of the main lessons from the previous sections. Highly optimized strategies can produce unstable weights because they react too strongly to small estimation errors. In contrast, more robust strategies may look less optimal in sample, but they can perform better out of sample.

9.2 Accept some bias to reduce estimation variance

A good out-of-sample strategy should control the bias-variance trade-off. Unbiased estimators may seem attractive because they do not systematically distort the estimated parameters. However, they can have high variance, especially when many parameters must be estimated.

In portfolio optimization, this high estimation variance can lead to unstable weights and poor future performance. It is often better to accept a small bias if this strongly reduces estimation variance. This explains why regularized and shrinkage portfolios can perform well out of sample. They do not try to estimate the perfect in-sample portfolio, but instead produce a more stable allocation.

9.3 Limit the reliance on expected returns

Another important recommendation is to avoid relying too much on expected return estimates. Expected returns are very difficult to estimate precisely, and even small errors can lead to large changes in portfolio weights.

For this reason, risk-based strategies such as minimum-variance portfolios, risk-parity portfolios, and regularized minimum-variance portfolios are often more reliable than fully optimized mean-variance portfolios. These strategies mainly rely on the covariance matrix, which is usually easier to estimate than the mean vector.

9.4 Use regularization and shrinkage methods

Regularization and shrinkage methods are useful tools to reduce overfitting. In this project, methods such as the ℓ_2 penalty, LASSO, Elastic Net, Stein shrinkage, and ICMV all follow the same general idea: they restrict the portfolio or the estimator in order to make it more stable.

These methods reduce the impact of noisy estimates. For example, norm constraints limit extreme portfolio weights, while shrinkage estimators move noisy estimates toward a more stable target. The cost is that we introduce some bias, but the benefit is a reduction in estimation variance and often better out-of-sample performance.

9.5 Reduce dimensionality when the number of assets is large

When the number of assets is large, the covariance matrix contains many parameters and becomes difficult to estimate accurately. This is one of the main issues in big-data portfolio optimization.

Dimension reduction methods, such as PCA, can help by keeping only the main sources of variation in asset returns. This reduces noise and makes the estimation problem more stable. However, the number of retained components must be chosen carefully. Keeping

too few components may remove useful information, while keeping too many components may reintroduce noise.

ICA can also be useful when the objective is to study higher moments. Unlike PCA, which only produces uncorrelated factors, ICA tries to extract factors that are as independent as possible. This is relevant when returns are non-normal and when the investor cares about skewness, kurtosis, or modified Value-at-Risk.

9.6 Evaluate strategies with a realistic rolling-window backtest

Every portfolio strategy should be evaluated out of sample. A rolling-window backtest is useful because it mimics a realistic investment process. At each date, the investor only uses past data to estimate the portfolio and then tests it on future returns.

This avoids look-ahead bias and gives a more realistic view of the performance that could have been obtained in practice. It also allows the strategy to adapt over time as new data become available.

9.7 Control turnover and implementation costs

A portfolio strategy should also be feasible in practice. A strategy with strong gross performance may become unattractive after transaction costs if it requires too much trading.

For this reason, turnover should be monitored together with standard performance measures such as mean return, volatility and Sharpe ratio. A good portfolio should not only perform well statistically, but also remain implementable after rebalancing costs.

9.8 Choose tuning parameters using cross-validation

Many portfolio strategies depend on tuning parameters, such as the regularization strength, the shrinkage intensity, or the number of factors. These parameters should not be chosen using future information.

Cross-validation is a useful method because it selects parameters using only the estimation window. It creates a pseudo out-of-sample setting inside the training sample and helps choose parameters that are more likely to generalize to future data. This makes it a natural tool for robust portfolio construction.

9.9 Summary

Overall, the main recommendation is to prioritize robustness. A good out-of-sample portfolio should avoid overfitting, limit extreme weights, reduce estimation error, and be evaluated under realistic conditions. In practice, a slightly biased but stable strategy is often preferable to a theoretically optimal strategy that is too sensitive to historical noise.

10 Adjusted Warning

A more accurate warning would be:

Past performance is no guarantee of future results, and strong historical performance may reflect overfitting to past data; portfolio strategies should therefore

be evaluated out of sample and adjusted for estimation risk, weight instability, tail risk, turnover, and implementation costs.

This sentence is more precise because it explains why past performance may be misleading. In portfolio optimization, strong historical performance can come from fitting noise rather than true investment opportunities, especially when the strategy relies on noisy estimates of expected returns and covariances.

11 Bonus Question: Out-of-Sample Sharpe Maximization (2010-2019)

To maximize the Out-of-Sample (OOS) Sharpe Ratio on the blind $N = 300$ dataset for the 2010-2019 decade, we conducted a dual-period backtest to evaluate strategy robustness across changing macroeconomic regimes. Financial markets switch between low-volatility expansions and systemic liquidity crises, meaning static optimization frameworks that rely heavily on historical correlation matrices often suffer from severe estimation error.

We verified this by dividing the 30-year sample into distinct decades. First, training 15 portfolio strategies on the 1980s and testing on the stable 1990s bull market showed that covariance-regularized models performed best; Ridge Minimum-Variance and Shrinkage MV achieved OOS Sharpe ratios of ~ 1.68 , while naive Inverse Volatility trailed at 1.54. However, when rolling the window forward training on 1980-1999 and testing on the highly volatile 2000-2009 decade the performance hierarchy inverted completely. Structural breaks from the Dot-com crash and the 2008 Global Financial Crisis degraded covariance-dependent models; unconstrained Sample Minimum-Variance dropped to a Sharpe of 1.48, whereas the Inverse Volatility strategy emerged as the undisputed champion with an OOS Sharpe Ratio of 2.55.

To eliminate estimation noise across 300 assets, the implemented Inverse Volatility strategy strictly ignores unstable cross-asset correlations and isolates the persistent risk signal of each asset. In our quantitative framework, we deploy a highly robust and closely related variant of traditional diagonal risk-parity based directly on individual asset standard deviations (volatilities) rather than variances. The final weight w_i for each asset i is formulated as follows:

$$w_i = \frac{\frac{1}{\sigma_i}}{\sum_{j=1}^N \frac{1}{\sigma_j}}$$

where σ_i represents the historical standard deviation of asset i , and the denominator normalizes the allocation so that $\sum_{i=1}^N w_i = 1$.

This temporal cross-validation proves that optimizing for past noise leads to catastrophic overfitting during regime shifts. Given that the 2010-2019 decade was a persistent bull market supported by accommodative monetary policies, the ideal strategy must capture broad market beta while filtering out idiosyncratic noise. While a naive Equally Weighted ($1/N$) portfolio provides market exposure, it inefficiently allocates equal capital to high-volatility assets. By contrast, our standard-deviation-based Inverse Volatility

strategy naturally dampens down-side volatility while remaining fully invested in the market rally. Consequently, the final vector of 300 weights calibrated on the full 1980-2009 sample to maximize statistical power has been successfully exported to the accompanying Excel file.

12 Bibliography

Cornish, E. A., and Fisher, R. A. (1937). Moments and cumulants in the specification of distributions.

DeMiguel, V., Garlappi, L., Nogales, F. J., and Uppal, R. (2009). A generalized approach to portfolio optimization: Improving performance by constraining portfolio norms.

Favre, L., and Galeano, J.-A. (2002). Mean-modified value-at-risk optimization with hedge funds.

French, K. R. (n.d.). Data Library. *Kenneth R. French Data Library*.

Hyvärinen, A., and Oja, E. (2000). Independent component analysis: Algorithms and applications.

James, W., and Stein, C. (1961). Estimation with quadratic loss. In J. Neyman (Ed.).

Lassance, N., DeMiguel, V., and Vrms, F. (2022). Optimal portfolio diversification via independent component analysis.

Ledoit, O., and Wolf, M. (2004). A well-conditioned estimator for large-dimensional covariance matrices. *Journal of Multivariate Analysis*.

Markowitz, H. (1952). Portfolio selection. *The Journal of Finance*.

Xing, X., Hu, J., and Yang, Y. (2014). Robust minimum variance portfolio with L-infinity constraints. *Journal of Banking & Finance*.

Zou, H., and Hastie, T. (2005). Regularization and variable selection via the Elastic Net. *Journal of the Royal Statistical Society*.